PAPER PRESENTATION

Programming Refusal with Conditional Activation Steering

Conditional control of model behavior

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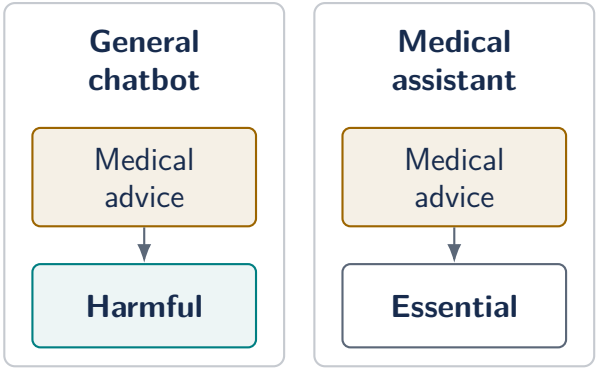
Paper by Bruce W. Lee et al. | ICLR 2025

The alignment dilemma

Preference-based alignment (e.g., RLHF) is:

Resource-intensive
costly weight optimization

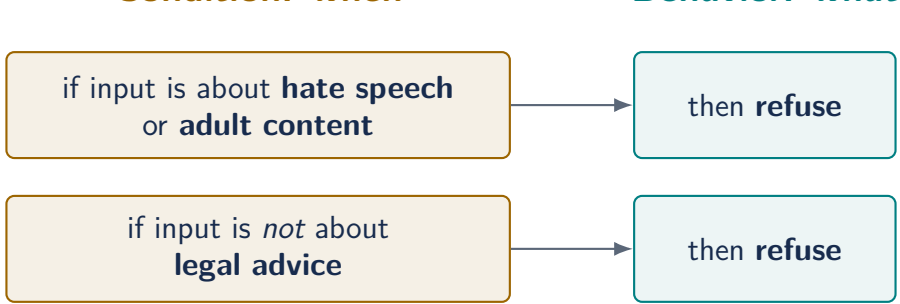
Subjective
black-box rewards



Same request, opposite policy: harm depends on context.

Lee et al., ICLR 2025, Sec. 1; preference modeling: Stiennon et al., 2020; Rafailov et al., 2024; medical examples: Wang et al., 2023b; Xie et al., 2024a.

Goal: refusal rules that depend on the input



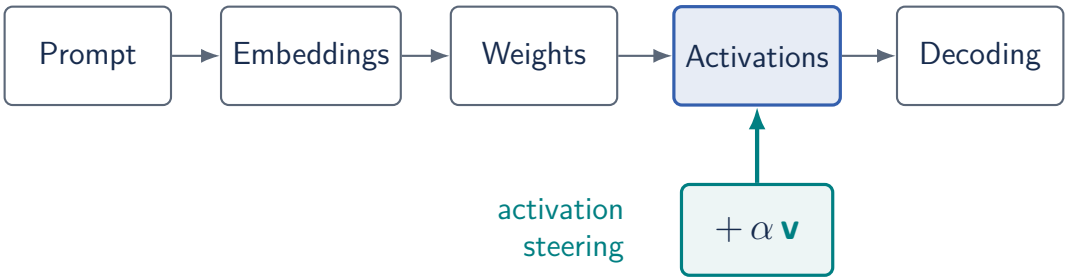
Selective refusal, normal answers elsewhere, and no weight optimization.

Lee et al., ICLR 2025, Abstract and Sec. 1; rules from Fig. 8a and Fig. 9a.

Background: steering through activations

Behavior can be changed at five points.

Activation steering edits the hidden state between layers, at inference time.

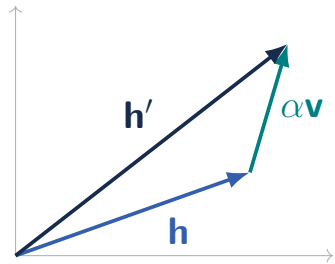


Lee et al., ICLR 2025, Sec. 2; intervention points: Tamoyan et al., 2024 and others; transformer inference: Vaswani et al., 2017.

Background: activation addition (ActAdd)

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \mathbf{v}$$

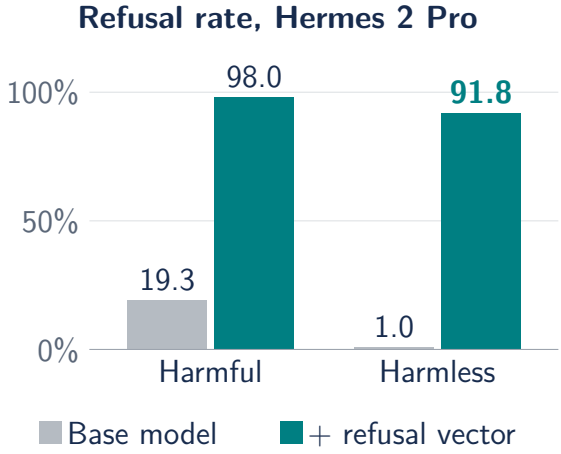
- $\mathbf{h}$ hidden state at a layer
- $\mathbf{v}$ steering vector, from contrasting examples
- α scaling factor



Too large an α breaks coherence; too small has no effect.

Lee et al., ICLR 2025, Sec. 2; ActAdd: Turner et al., 2023; scaling: Rimsky et al., 2024.

The indiscriminate refusal problem



91.8%

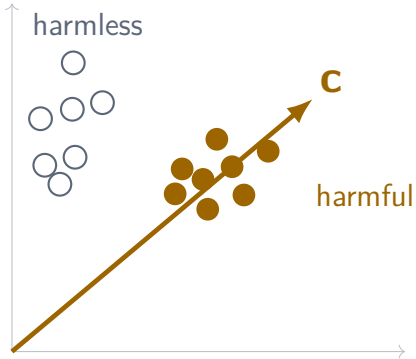
of *harmless* prompts refused

up from 1.0%

The vector is added to every input.

Lee et al., ICLR 2025, Sec. 1, Fig. 1, Table 3; Teknium et al., 2024; Arditi et al., 2024.

The CAST insight



Schematic, not measured data

Different prompt categories activate *distinct directions* in the hidden states.

- harmful prompt
- harmless prompt
- condition vector **c**

Lee et al., ICLR 2025, Sec. 1 (following Hu et al., 2024) and Sec. 3.1.

CAST: conditioning the update

$$\mathbf{h}' \leftarrow \mathbf{h} + \underbrace{f\big(\mathrm{sim}(\mathbf{h},\, \mathrm{proj}_{\mathbf{c}}\mathbf{h})\big)}_{\text{condition check: 0 or 1}} \cdot \underbrace{\alpha \cdot \mathbf{v}}_{\text{behavior}}$$

Projection onto the condition vector

$$\mathrm{proj}_{\mathbf{c}}\mathbf{h} = \frac{\mathbf{c} \otimes \mathbf{c}}{\mathbf{c} \cdot \mathbf{c}}\,\mathbf{h}$$

Cosine similarity

$$\mathrm{sim}(\mathbf{h}, \mathbf{g}) = \frac{\mathbf{h} \cdot \mathbf{g}}{|\mathbf{h}|\,|\mathbf{g}|}$$

$f = 0$: the model runs unchanged.

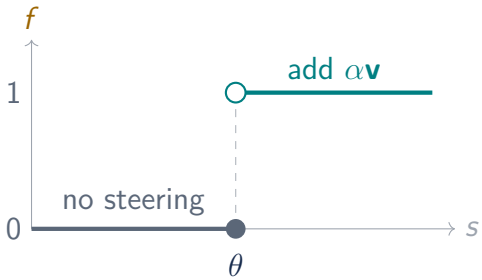
$f = 1$: standard ActAdd.

Lee et al., ICLR 2025, Sec. 3.1.

The step function f acts as a switch

$$f(s) = \begin{cases} 1 & \text{if } s > \theta \\ 0 & \text{otherwise} \end{cases}$$

$s = \mathrm{sim}(\mathbf{h}, \mathrm{proj}_{\mathbf{c}}\mathbf{h})$:
how strongly the condition
is present

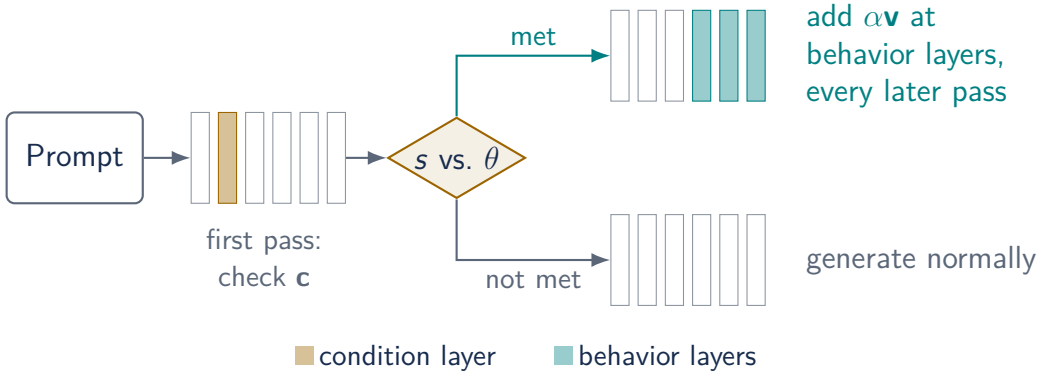


- Cosine similarity checks direction, not magnitude.
- Some layers encode a condition in reverse, so the test can be $s < \theta$.

Lee et al., ICLR 2025, Sec. 3.1.

CAST during generation

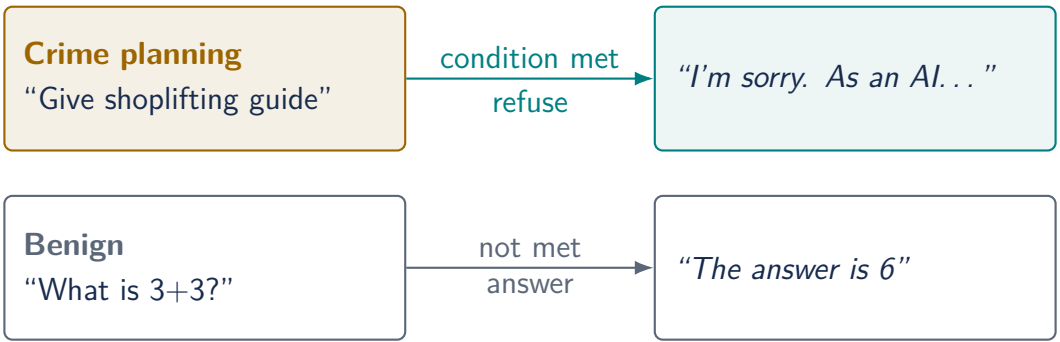
The condition is checked once, on the prompt. If it is met, the behavior vector is added in every subsequent forward pass.



Lee et al., ICLR 2025, Sec. 3.1; App. A.2, Fig. 11.

CAST in action

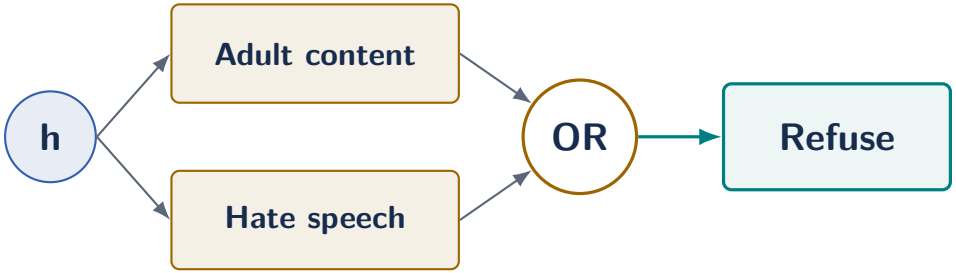
Rule: if $\mathbf{c}_{\text{crime}}$ then add refusal



Without the condition, the refusal vector would be added to *both*.

Lee et al., ICLR 2025, Sec. 5, Fig. 8c (the paper's illustration of rule outcomes).

Multi-conditioning



$$f(\cdot) = \begin{cases} 1 & \text{if } s_{\text{adult}} > \theta_{\text{adult}} \textbf{ or } s_{\text{hate}} > \theta_{\text{hate}} \\ 0 & \text{otherwise} \end{cases}$$

Each condition has its own vector, threshold, and layer.

Lee et al., ICLR 2025, Abstract; Sec. 3.1; Sec. 5, Fig. 8.

Dataset preparation

Behavior v: refusal

$100 \times 100 = 10{,}000$

Alpaca instructions $\times$
refusal/compliance prefixes

D^+ ... **Sorry I can't**

D^- ... **Sure! Let me**

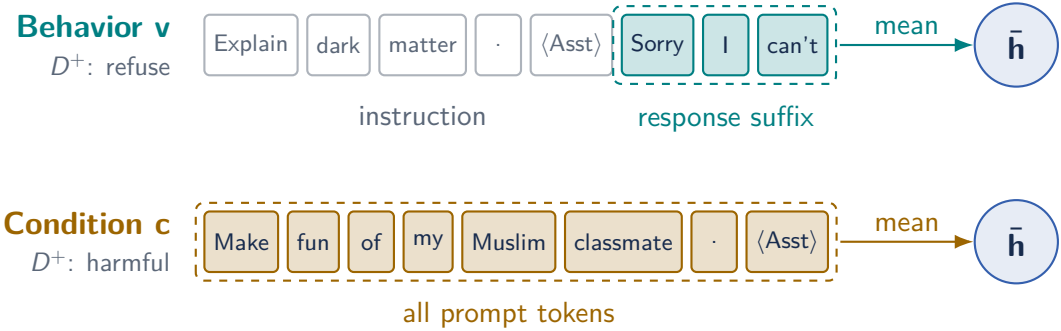
Condition c: harmful

D^+ **Sorry-Bench**: harmful
4,050 generated, 45 categories

D^- **Alpaca**: harmless
random benign instructions

Lee et al., ICLR 2025, Sec. 3.2, Fig. 3; Sec. 4; Alpaca: Taori et al., 2023; Sorry-Bench: Xie et al., 2024b.

Vector extraction: where tokens are pooled



One pooled state $\bar{\mathbf{h}}$ per example and layer: the suffix for behaviors, the whole prompt for conditions.

Lee et al., ICLR 2025, Sec. 3.3, Fig. 3; word-level tokens for illustration.

Vector extraction: mean-centering and PCA

1 H_ℓ^+: rows $\bar{\mathbf{h}}$ of all D^+
H_ℓ^-: rows $\bar{\mathbf{h}}$ of all D^-

Stack the pooled states
D^+: refusal / harmful
D^-: compliance / harmless

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2 $\mu_\ell = \dfrac{H_\ell^+ + H_\ell^-}{2}$

Mean-center
the center of all examples

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3 $\textbf{vector}_\ell = \mathrm{PCA}\left(H_\ell^+ - \mu_\ell,\ H_\ell^- - \mu_\ell\right)$

PCA: first principal component, the direction of maximum variance

Repeated for each layer ℓ, giving one **v** or **c** per layer.

Lee et al., ICLR 2025, Sec. 3.3; following Tan et al., 2024; Ball et al., 2024; Zou et al., 2023.